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The Effect of Peer Contact Upon Physiological Response to Maternal Separation

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The response of both human and nonhuman primate infants to separation from their mothers is characterized by agitation or protest behaviors followed by a behavioral depressive reaction. In monkeys, this behavioral syndrome has striking physiological correlates, including heart rate and sleep disturbances. There is, however, considerable individual variability in response to maternal separation. The authors hypothesized that the amount of contact with particular, familiar conspecifics might account for some of this variability. In support of this notion, data are presented that indicate that pigtailed monkey infants who have been separated from their mothers but who maintain relatively high levels of peer contact do not show as severe decreases in heart rate during maternal separation as do those that exhibit less peer contact.

Key words: attachment, peer bonds, maternal separation, depression, biotelemetry, heart rate, macaques

INTRODUCTION

Disruption of attachment bonds can lead to severe behavioral and physiological disturbances in many higher organisms. In human infants, loss of one's mother leads to a syndrome described by Spitz [1946] as "anaclitic depression." Similarly, many nonhuman primate infants exhibit "agitation-depression" or "protest-despair" reactions when confronted with the loss of their mothers [Hinde & Spencer-Booth, 1971; Seay et al, 1962; Kaufman & Rosenblum, 1967]. Recently, the physiological concomitants of the behavioral expression of the separation syndrome have been brought to light by Reite and his coworkers [Reite et al, 1978b; Reite & Short, 1978; Reite et al, in press]. These studies have shown, for example, that the initial agitation reaction to separation (which occurs during the first 24 hours following separation) is accompanied by increases in heart rate and body temperature, as well as by sleep disturbances (eg, decreased amounts of REM sleep). The depressive or withdrawal stage which follows the initial reaction is characterized by decreases in body temperature and heart rate, along with continued sleep disturbances.

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Summarizing the "typical" responses to maternal separation does not, however, do justice to the great range of individual differences shown by infant primates. Suomi [1976] states that "there is no universal reaction to social separation in monkeys," and Reite et al [in press] remark that "the individual variability in response to separation was remarkable." Suomi et all [in press] suggest that differences in social relationships prior to separation, variation in separation environment, and postseparation variables are probably responsible for observed individual differences, but they also note that "there has not been much work directly investigating the relative contribution of these different sources of variance, nor has there been much effort designed to see how such factors might interact with each other in various mother-infant separations." The theoretical and clinical relevance of determining the extent to which the separation response is exacerbated, mitigated, or otherwise mediated by biological, psychological, and sociological factors is evident. Henderson [1980], for example, has asserted that "it is now necessary to ... find out what deficiencies in social relationships are most associated with specific psychiatric diagnoses." In the realm of separation-induced depression, Bowlby [1980] has emphasized the need for continued study of the syndrome in order to develop effective measures for treatment and/or prevention.

Intuitively, it is reasonable to assume that responses to maternal separation would be affected by the nonmaternal attachments within the infant's social sphere. Bonnet macaque infants, for example, do not become depressed upon being separated from their mothers as long as the former are allowed to maintain contact with others in their typically cohesive social groups [Kaufman & Stynes, 1978]. Cross-cultural studies have presented evidence that, in those human societies where kin and friends provide extensive (and often ritualized) support for the bereaved, disabling grief syndromes rarely occur [eg, Ablon, 1971]. Nevertheless, systematic studies of the extent to which peer attachments influence a child's response to social loss have not been pursued. Indeed, normative studies of human peer attachments and response to peer separation are conspicuous only for their absence. This is a curious dearth, especially considering the anecdotal evidence in favor of the notion that peer attachments and separations can have significant effects on children [eg, see Hartup, 1975].

The nonhuman primate literature provides ample evidence for the importance of peer attachments and the debilitating effects of peer separation [eg, Suomi et al, 1970], but most of this work has not been carried out in the context of a "natural" social setting. We therefore know virtually nothing about the response of an immature primate living in a social network including its mother, other adults, and other peers, to the loss of a particular peer. Neither do we know much about the role of peer bonds in contributing to an individual's response to other forms of social loss [but see Gunnar et al, 1980].

The results presented below are, to the best of our knowledge, the first data to support the notion that peer relationships among nonhuman primates affect an infant monkey's physiological response to the disruption of an attachment bond. These data suggest that spending increased amounts of time with peers can mitigate the physiological response to maternal separation. Although the small number of animals makes these conclusions tentative, we feel that the results warrant exposure, as well as further investigation.

METHODS

Our subjects were six infant pigtailed macaques (*Macaca nemestrina*). All were born in the Department of Psychiatry Primate Laboratory, and were raised by their mothers in social groups containing four to eight adult females, some with offspring, and one adult male. The pens, which were equipped with one-way windows for observation, were 2.1 × 2.5 × 4 meters.

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Between 20 and 30 weeks of age, each infant was surgically implanted with biotelemetry systems that transmit multivariable physiological data including heart rate (HR), body temperature (BT), eye movement, muscle activity, and three channels of EEG [Pauley & Reite, in press]. Infants were returned to their mothers immediately after recovery from anesthesia. Sutures were removed approximately 1 week following surgery, and baseline behavioral and physiological data collection started between 12 days and 6 weeks after implantation (a detailed description of behavioral and physiological data collection is presented in Reite and Short [1980]). Following 5–12 baseline days, the mothers were removed, and the infants remained in the otherwise intact social groups. After 10–12 days of maternal separation, the mothers were returned to the groups. Behavioral and physiological data collection continued uninterrupted.

Five-minute behavioral observations were made six times daily on each of the infants (designated as "focus" animals while being observed). A trained observer watched the focus animal for a 5-minute period and entered all behaviors, as they occurred, on a computer video terminal [Walker & Reite, 1978]). The behavioral taxonomy used is described in Reite and Short [1980]. Absolute start and stop times of all behaviors were stored, along with temporal sequence, and the duration of each behavior noted was summarized in terms of the percent of total time observed (% TTO). Results from single observation sessions were averaged to obtain mean daily scores for each of the behavioral measures. We will discuss the peer contact score in the present paper. Contact was operationally defined as any "passive" physical contact with another animal (other than the mother) but did not include aggression or play. Because the behavioral scoring program permitted the identification of the specific animal with which the focus animal was in contact, contact with peers was distinguishable from contact with other females or with the male.

Discrete HR measurements were obtained by the technician in attendance every 20 minutes during the day and every 15 minutes during the night. All HR samples obtained between 1000 and 1600 hours were averaged to obtain a measure called "mean day HR."

The contact scores and mean day HR values during the separation period were transformed into percent change from baseline scores. Percent change (%△) was computed by subtracting the baseline mean from the mean separation score and dividing the difference by the baseline mean. Percent △ scores were computed for contact and HR scores on separation day 1 (S_1), which corresponds to the agitation phase of separation, and for the contact and HR data averaged over the remainder of the separation period ($S_2 - S_x$). Contact and heart rate scores over $S_2 - S_x$ showed relatively little variability and were, therefore, averaged to aid and expediate analysis and interpretation. The $S_2 - S_x$ days correspond to the period accompanied by behavioral depression. Kendall's rank correlation coefficient (τ) was used to determine the extent to which increased amounts of contact with other animals was correlated with HR change.

RESULTS

The changes in peer contact and HR during separation are listed in Table I. On the S_1 day, HR increased in five of the six subjects; peer contact increased in three, and decreased in three. There was a positive correlation between HR change and peer contact change ($\tau = 0.60$, $p = 0.068$), suggesting that those animals that experienced the greatest HR increases also exhibited increased levels of peer contact. During the subsequent period of behavioral depression ($S_2 - S_x$), generally accompanied by precipitous HR drops, HR did indeed dip below baseline means in five of the six subjects. During this time there was again a positive correlation between HR change and change in peer contact scores ($\tau = 0.73$, $p = 0.028$); that is, those infants that maintained above-baseline

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TABLE I. Animal Identification Number, Sex, Date of Birth in Julian Day and Year, Age at Separation (in Days), and Change in Peer Contact and Daytime Heart Rate Scores as Measured by Percent Change From Baseline Values

Animal ID	Sex	DOB	Age at separation	Contact		HR	
				% △ S_1	% △ $S_2 - S_x$	% △ S_1	% △ $S_2 - S_x$
11.6	F	189-77	154	+128	+155	+20.0	+11.2
23.4	F	237-78	193	+112	− 66	+ 8.0	−40.3
21.2	M	74-78	228	+ 36	+ 36	+ 0.4	− 8.8
19.2	M	353-77	174	+ 18	+ 10	+ 5.0	−10.6
16.4	F	111-77	147	− 43	− 4	+ 4.2	−11.8
18.4	M	331-79	250	− 29	+ 28	− 5.3	−22.6

See text for details.

levels of peer contact were least susceptible to the HR declines associated with the depressive stage of separation from mother.

DISCUSSION

These data indicate that the one physiological response to maternal separation, HR, is strongly correlated with peer contact behavior. During the S_1 period, the period of behavioral agitation immediately following separation, the fact that most infants exhibited both increased peer contact and increased HR suggests that those infants most upset initially actively sought the "reassurance" of peers, as reflected in increased peer contact scores. Candland and Mason [1968] and Mason [1967] have argued that monkeys seek arousal reduction via "calm" sorts of contact with conspecifics; the physical aspects of the contact observed in our separated infants may have served this same function. In support of this notion, Reite et al [in press] have shown that the frequency of response to play (an arousal-inducing activity) is markedly decreased in separated infants. Furthermore, while the temporal (sequential) relationship between HR and peer contact changes was not quantified, our data did show that the infants' HR increased virtually immediately upon removal of their mothers, whereas, generally speaking, it was at least several hours before the infants, which were extremely agitated during this period, showed any interest in establishing contact with their group mates. High arousal did, therefore, precede contact. However, physically mediated arousal reduction is probably not the complete explanation for contact seeking in these animals: Reite et al [1978a] have shown that mere physical contact (gained through adoption by an adult female) was not sufficient to allay a physiological response to maternal separation in a pigtailed infant. Additionally, body temperature showed aberrations from normal even in those infants that maintained peer contact during separation. It thus appears that the psychological functions of contact, whatever they may be, are of great importance to the stressed infant. The continued correlational pattern throughout the period of behavioral depression suggests that maintenance of this social support over the course of separation may have reduced the magnitude of the HR decrease typically associated with the depressive phases of the separation reaction. Of course, an equally plausible explanation would be that those infants which, for whatever reason, were prone to greater drops in HR following separation, possibly indicating a greater degree of physiological disorganization, would be less able to maintain close peer bonds.

There are many questions raised by the data presented above. For example, what were the natures of the relationships which, apparently, provided a source of comfort for the temporarily orphaned infants? What are the qualities of a relationship that mitigate a response to loss? What are the effects of gender, age, and social rank on the role of the

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peer network in buffering separation stress? To what extent are the behavioral correlates of the depressive reaction influenced by peer support? Clearly, intercorrelations between more of the behavioral and physiological measures would paint a more complete picture of the relationship between agitation-depression and social bonds with peers. However, given the small sample size and the inherently variable nature of these sorts of behavioral and physiological measures, it seemed most prudent to delay such analyses until such time as a larger population of subjects will allow for the determination of further correlations, patterns, and trends. However, we believe that the strength of the HR–peer contact correlation is impressive when one considers the small sample size and the wide range of heart rate responses. We are therefore optimistic regarding the significance of future analyses, and it is the goal of our research program to identify the roles played by various social agents in the response to loss. It is our further hope that the derived knowledge can be implemented in therapeutic regimes for humans faced with the effects of loss and that it will ultimately contribute to an improved understanding of the pathophysiology of grief [Reite, 1979].

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REFERENCES

Ablon, J. Bereavement in a Samoan community. BRITISH JOURNAL OF MEDICAL PSYCHOLOGY 44:329-337, 1971.

Bowlby, J. MATERNAL AND MENTAL HEALTH. World Health Organization, Monograph Series No. 2, 1952.

Caine, N. The development of peer preferences in immature macaques. PhD Dissertation, University of California, Davis, 1980.

Candland, D.K.; Mason, W.A. Infant monkey heart rate: Habituation and effects of social substitutes. DEVELOPMENTAL PSYCHOBIOLOGY 1(4): 254-256, 1968.

Gunnar, M.R.; Gonzales, C.A.; Levine, S. The role of peers in modifying behavioral distress and pituitary-adrenal response to a novel environment in year-old rhesus monkeys. PHYSIOLOGY AND BEHAVIOR 25:795-798, 1980.

Hamburg, D.A.; Hamburg, B.A.; Barchas, J.D. Anger and depression in the perspective of behavioral biology. pp. 300-317 in EMOTIONS – THEIR PARAMETERS AND MEASUREMENT. L. Levi, ed. New York, Raven Press, 1975.

Hartup, W.W. The origins of friendship. pp. 11-26 in FRIENDSHIP AND PEER RELATIONS. M. Lewis, L. Rosenblum, eds. New York, Wiley & Sons, 1975.

Henderson, S. A development in social psychiatry: a systematic study of social bonds. JOURNAL OF NERVOUS AND MENTAL DISEASES 168:63-69, 1980.

Henoch, M.J.; Batson, J.W.; Baum, J. Psychosocial factors in juvenile rheumatoid arthritis. ARTHRITIS AND RHEUMATISM 21(2): 229-233, 1978.

Hinde, R.A., Spencer-Booth, Y.: Effects of brief separation from mother on rhesus monkeys. SCIENCE 173:111-118, 1971.

Jacobs, T.J.; Charles, E. Life events and the occurrence of cancer in children. PSYCHOSOMATIC MEDICINE 42:11-24, 1980.

Kaufman, I.C.; Rosenblum, L. Effects of separation from mother on the emotional behavior of infant monkeys. ANNALS OF THE NEW YORK ACADEMY OF SCIENCE 159:681-695, 1969.

Kaufman, I.C.; Stynes, A.J. Depression can be induced in a bonnet macaque infant. PSYCHOSOMATIC MEDICINE 40:71-75, 1978.

Mason, W.A. Motivational factors in psychosocial development. pp. 35-67 in NEBRASKA SYMPOSIUM ON MOTIVATION. W.J. Arnold; M.M. Page, eds. Lincoln, University of Nebraska Press, 1971.

McKnew, D.H.; Cytryn, L. Historical background in children with affective disorders. AMERICAN JOURNAL OF PSYCHIATRY 130:1278-1280, 1973

Morillo, E.; Gardner, L.I. Bereavement as an antecedent factor in thyrotoxicosis of childhood: Four case studies with survey of possible metabolic pathways. PSYCHOSOMATIC MEDICINE 41:545-555, 1979.

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Pauley, J.D.; Reite, M. A microminiature hybrid multichannel implantable biotelemetry system. BIOTELEMETRY AND PATIENT MONITORING (in press).

Reite, M. Towards a pathophysiology of grief. PSYCHOSOMATIC MEDICINE 41:78, 1979 (abstract).

Reite, M.; Seiler, C.; Crowley, T.J.; Hydinger-Macdonald, M.; Short, R. Circadian rhythm changes following maternal separation in monkeys. (unpublished manuscript).

Reite, M.; Seiler, C.; Short, R. Loss of your mother is more than loss of a mother. AMERICAN JOURNAL OF PSYCHIATRY 135:370-371, 1978a.

Reite, M.; Short, R.A. Nocturnal sleep in separated monkey infants. ARCHIVES OF GENERAL PSYCHIATRY 35:1247-1253, 1978.

Reite, M.; Short, R. A biohavioral developmental profile (BDP) for the pigtailed monkey. DEVELOPMENTAL PSYCHOBIOLOGY 13:243-285, 1980.

Reite, M.; Short, R.; Kaufman, I.C., Stynes, A.J.; Pauley, J.D. Heart rate and body temperature in separated monkey infants. BIOLOGICAL PSYCHIATRY 13:91-105, 1978b.

Reite, M.; Short, R.; Seiler, C.; Pauley, J.D. Attachment, loss and depression. JOURNAL OF CHILD PSYCHOLOGY AND PSYCHIATRY (in press).

Seay, B.; Hansen, E.W.; Harlow, H.F. Mother-infant separation in monkeys. JOURNAL OF CHILD PSYCHOLOGY AND PSYCHIATRY 3:123-132, 1962.

Seiler, C.; Cullen, J.S.; Zimmerman, J.; Reite, M. Cardiac arrhythmias in infant pigtail monkeys following maternal separation. PSYCHOPHYSIOLOGY 16:130-135, 1979.

Singh, S.D. Effects of infant-infant separation of young monkeys in a free-ranging natural environment. PRIMATES 18:205-214, 1977.

Spitz, R.A. Anaclitic depression. PSYCHOANALYTIC STUDY OF THE CHILD 2:313-342, 1946.

Soumi, S.J. Factors affecting responses to social separation in rhesus monkeys. MODELS IN HUMAN PSYCHOBIOLOGY. G. Serban,; A. Kling,; eds. New York, Plenum, pp. 9-26, 1976.

Suomi, S.; Harlow, H.; Domek, C. Effect of repetitive infant-infant separation of young monkeys. JOURNAL OF ABNORMAL PSYCHOLOGY 76(2):161-172, 1970.

Soumi, S.J.; Mineka, S.; Harlow, H.F. Social separation in monkeys as viewed from several motivational perspectives. In HANDBOOK OF BEHAVIORAL NEUROLOGY: MOTIVATION. P. Teitelbaum, E. Satinoff, eds. New York, Plenum (in press).

Walker, S.D.; Reite, M. BEIN11: On-line behavior input. PROCEEDINGS OF THE DIGITAL EQUIPMENT USERS SOCIETY, San Francisco, November 1978, pp. 715-717, DECUS, Maynard, MA 01754.

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